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Education

2019 **B.A. Computer Science and Math**, UMass, Amherst

Professional Experience

Rust, Haskell, TypeScript/JavaScript, Cryptography, Distributed Systems, Web Servers, Internet Standards, Security, PostgreSQL, Kubernetes, Smart Contracts

2022-now **Senior Software Engineer**, *Google*, Remote

- Lead a team developing anonymous credentials protocols for the web using zero-knowledge proofs, with implementation in Rust.
- Published open source anonymous credentials implementations in the [authenticated-pseudonyms](#) and [anonymous-credit-tokens](#) libraries.
- Reviewed web proposals for privacy and defensibility properties.
- Designed the [Private Proof API](#) as a solution for anti-fraud providers in a web without third party cookies, leading a group of three web platform engineers to implement it in Chrome.
- Active in the W3C
 - Chair of the Anti-Fraud Community Group
 - Federated Identity Working Group
 - Security Interest Group
 - Verifiable Credentials Working Group
- Active in IETF
 - Privacy Pass Working Group
 - Crypto Forum

2022 **Software Engineer / Scientist**, *Casper Labs*, Remote

- Co-designed a novel consensus algorithm: [preprint](#). This work was presented at FOCODILE 2022.
- Implemented this consensus algorithm in the [Casper reference node](#). The algorithm was recently deployed as a part of the 2.0 upgrade.
- Conducted research on zero-knowledge use cases in the Casper ecosystem. Proposed changes to optimize the network for zero-knowledge use cases.
- As part of the research team, consulted for the C-suite on strategy for the network's design from a security and scalability perspective.

2019-2022 **Software Engineer (Team Lead)**, *SimSpace*, Boston/Remote

- Designed and implemented Attack Designer program.
- Designed consensus-lite protocol for multiple data-center clusters to coordinate across each other.
- As backend security lead, found and mitigated several incorrect cryptographic applications.